

Vansh Agrawal

 [vansh070605](#) |  [thevanshagrawal](#) |  vansh070605@gmail.com |  +91 88102 23022

B.Tech, Computer Science & Engineering (AI & ML) | SRM IST, Kattankulathur | Graduating 2027

SUMMARY

Final-year CS student and AI/ML Engineer specializing in **computer vision**, **multi-agent RAG architectures**, and **real-time edge intelligence**. Currently an AI/ML Intern at **Tata Steel**, building scalable CV solutions for industrial quality control. Focused on deployable, **low-latency**, production-grade ML systems across autonomous agent frameworks and real-time vision.

SKILLS

Languages	Python , JavaScript, TypeScript, SQL , Java, C/C++, HTML5/CSS3
Frameworks & Libraries	TensorFlow , OpenCV , Google MediaPipe, SentenceTransformers , FastAPI , React 19, Scikit-learn, Pandas, NumPy
Tools & Databases	Git/GitHub, Docker, ChromaDB (Vector DB), Firebase, PostgreSQL , MySQL, Vercel, REST APIs

WORK EXPERIENCE

Tata Steel

Nov 2025 – Present

AI/ML Intern — Chennai, India

- Engineered a **Computer Vision** pipeline using **OpenCV** for automated TMT rebar quality control, calculating rib parameters with **95% precision** to evaluate pass/fail tolerances.
- Developed a validation module for chemical ring tests using **spatial analysis** to measure rod thickness and reduce manual inspection time by **40%**.

Digital Info Solutions Pvt Ltd

Dec 2024 – Jan 2025

Internship Trainee — Ghaziabad, Uttar Pradesh, India

- Developed **full-stack web applications** using **Flask**, integrating Python, HTML, CSS, JavaScript, and **MySQL**.
- Designed and implemented **backend functionalities**, database connectivity, and responsive user interfaces through hands-on project development.

PROJECTS

PlacementOS — AI Career Command Center

 [GitHub](#)

React 19, FastAPI, ChromaDB, SentenceTransformers, TensorFlow, Firebase

- Local-first AI career platform with **9 autonomous agents**; **RAG loop** using **SentenceTransformers** and **ChromaDB** to extract JD metrics and tailor resumes via the Google X-Y-Z formula.
- Built a career compass, project defense Q&A generator, and a **TensorFlow/Keras regression model** for localized salary band prediction.

WakeGuard — AI Driver Safety Vision System

 [GitHub](#)

React, Vite, Google MediaPipe, Tailwind CSS, Recharts

- Real-time client-side driver monitoring tracking **478 3D facial landmarks at 60 FPS** using **MediaPipe Face Mesh**.
- Computes **EAR**, **MAR**, and **3D head rotation** to detect microsleeps, yawning, and distraction; voice alerts via **Web Speech API** with zero backend dependencies.

AERIS — Flood Risk Intelligence System

 [GitHub](#)

XGBoost, Python, FastAPI, React 19, GeoJSON

- AI-powered flood management platform for Bihar with proactive **7-day disaster response intelligence** using a **FastAPI** backend and React/Tailwind frontend.
- Dual XGBoost models** and a **React-Leaflet GIS dashboard** analyzing live telemetry across **18 river stations** with automated breach predictions.

EDUCATION

SRM IST, Kattankulathur	B.Tech — CSE (AI & ML) CGPA: 7.86 / 10	2023 – 2027
A.P.J School, NOIDA	Class XII — CBSE (MPC) 68.4%	2023
A.P.J School, NOIDA	Class X — CBSE 85.2%	2021